

CS224n & Transformer

2026.04.30

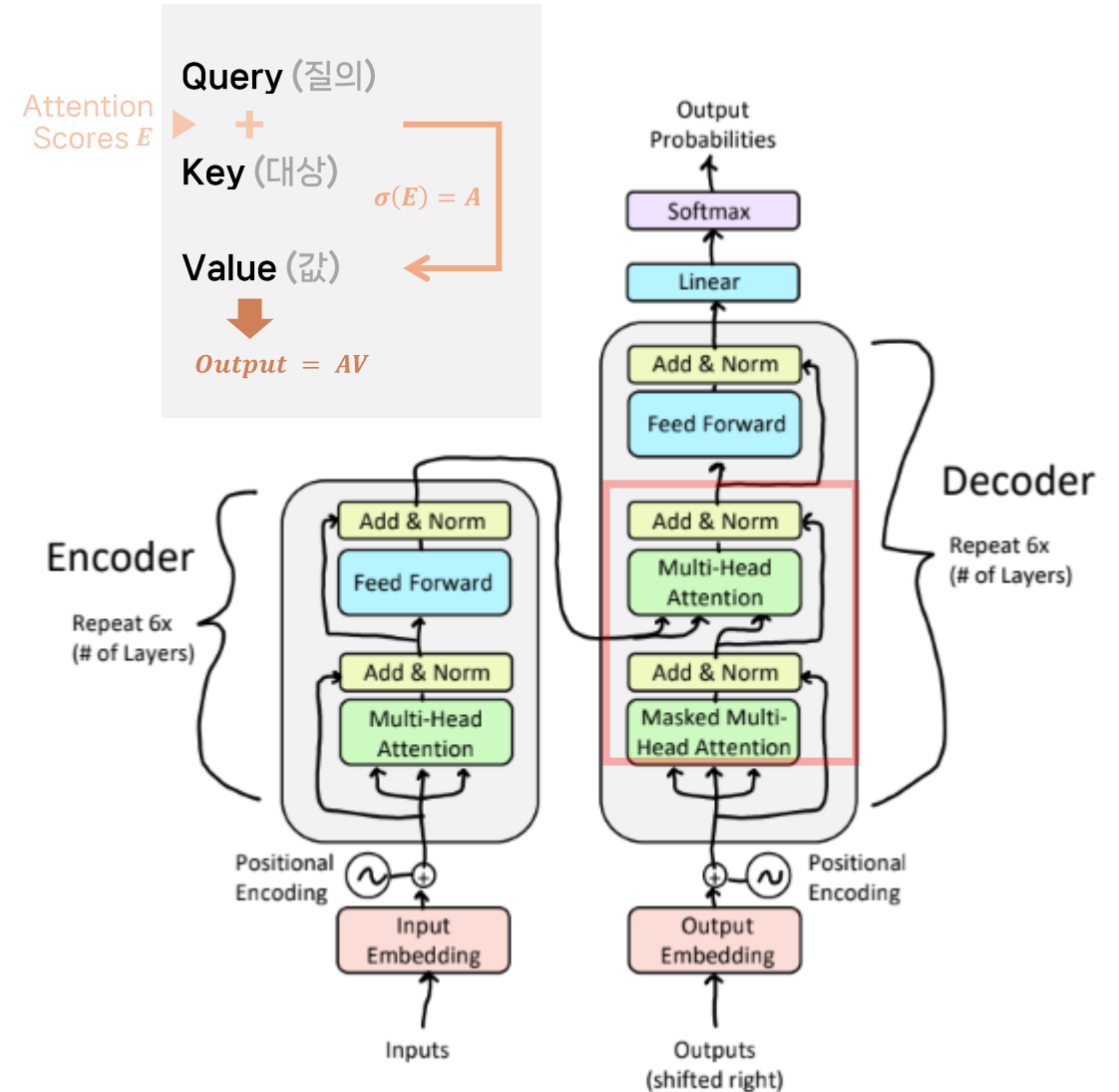
Hayeong Jung

Contents

- **Review**
 - Lecture 9
- **CS224n**
 - Lecture 10: Post-training
 - Lecture 11: Benchmarking & Evaluation
- **Paper Review**
 - Attention is All You Need(2017)

Review

- **Attention vs Self-Attention**
 - Attention from decoder to Encoder
 - Self-Attention in Decoder or Encoder
- **Transformer**
 - Encoder
 - Multi-Head (Self) Attention
 - Residual Connection
 - Layer Normalization
 - Positional Encoding
 - Decoder
 - Like Encoder;
 - Masked Attention
 - Decoder-Encoder Attention



Lecture 10: Post-training

• What We Learn from Pretraining?

- Trivia
- Syntax
- Coreference
- Lexical Semantics
- Sentiment
- Reasoning
- Arithmetic...

Truth-based
Token Prediction

⇒ Can do modeling of
More **Complex Concept/System**

Stanford University is located in _____, California. [Trivia]

I put ___ fork down on the table. [syntax]

The woman walked across the street, checking for traffic over ___ shoulder. [coreference]

I went to the ocean to see the fish, turtles, seals, and _____. [lexical semantics/topic]

Overall, the value I got from the two hours watching it was the sum total of the popcorn and the drink. The movie was _____. [sentiment]

Iroh went into the kitchen to make some tea. Standing next to Iroh, Zuko pondered his destiny. Zuko left the _____. [some reasoning – this is harder]

I was thinking about the sequence that goes 1, 1, 2, 3, 5, 8, 13, 21, _____. [some basic arithmetic; they don't learn the Fibonacci sequence]

• Emergent Ability

- Zero-Shot Learning
- Few-Shot Learning
 - In-context Learning
 - w/o Gradient Updates

The Difference

New methods of “prompting” LMs

Zero/few-shot prompting

1 Translate English to French: ←
2 sea otter => loutre de mer ←
3 peppermint => menthe poivrée ←
4 plush girafe => girafe peluche ←
5 cheese => ←

Traditional fine-tuning



Lecture 10

- Chain-of-thought Prompting

- Give Reasoning Example

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

- Zero-shot CoT

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

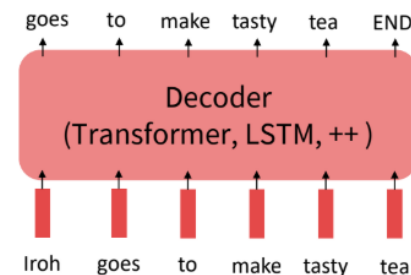
A: **Let's think step by step.** ◀ Make the model think by itself

- Finetuning & Instruction Finetuning

- Limits to Alignment
 - Language Model $\neq$ User Intent
 - Instruction Finetuning
 - Lots of Instruction-Output Pairs
 - Contains Various Tasks(' Context)
 - Evaluate on Unseen Tasks

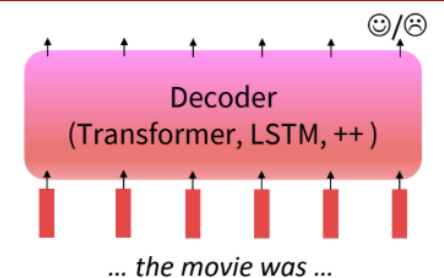
Step 1: Pretrain (on language modeling)

Lots of text; learn general things!



Step 2: Finetune (on **many** tasks)

~~Not~~ many labels; adapt to the tasks!



Lecture 10

- Limitation of Instruction Finetuning

- Expensive
- Open-Ended Problem
 - There's No 'One' Right Answer
- Same Penalties on Mistakes
 - Some mistakes can be worse than others
- Imperfectness of Human
 - Can't Always Generate the Optimal Answer

⇒ **How to Explicitly/Directly Satisfy Human Preference?**

- RLHF

- Reinforcement Learning from Human Feedback

- $R(x, y_i) = \mathbb{R}$

- R : **Reward Model**

- x : Input (of Specific Task)

- y_i : i^{th} Output (of Specific Task)

⇒ Large Reward Score is Better

- How to get Reward Model?

- Human Ratings(Scorings) are Expensive & Unreliable(variability)

- Use a **Pairwise Comparison**

⇒ **Bradley-Terry Pair Comparison Model**

$$\mathcal{L}_{\mathcal{RM}}(\phi) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathbb{D}} \left[\log \sigma \left(r_{\phi}(x, y_w) - r_{\phi}(x, y_l) \right) \right]$$

y_w : Winning Output
 y_l : Losing Output

y_w should be Larger than y_l to Minimize the Loss

Lecture 10

• RLHF: Reward Model

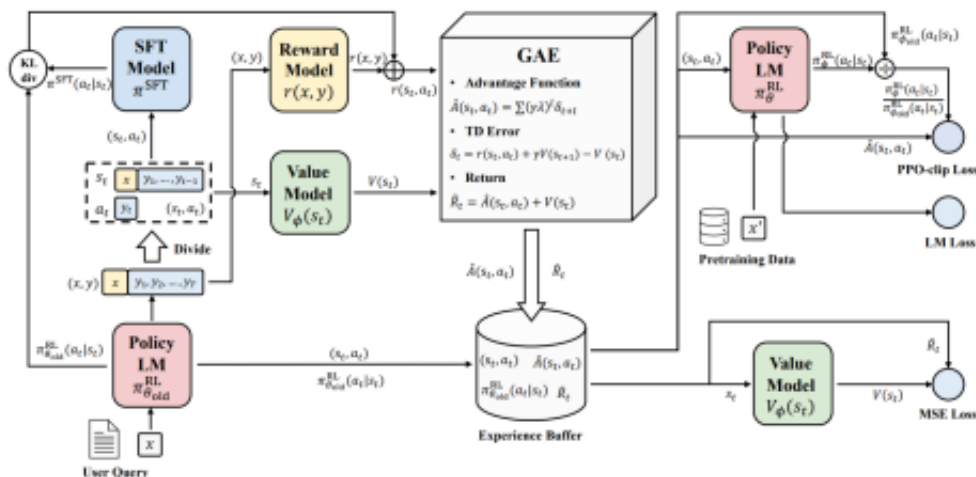
- Add Penalty to Reward Model

$$\begin{aligned} & \mathbb{E}_{\hat{y} \sim p_{\theta}^{\text{RL}}(\hat{y}|\mathbf{x})} [\mathbf{RM}_{\phi}(\mathbf{x}, \hat{y})] \\ & \mathbb{E}_{\hat{y} \sim p_{\theta}^{\text{RL}}(\hat{y}|\mathbf{x})} \left[\mathbf{RM}_{\phi}(\mathbf{x}, \hat{y}) - \beta \log \left(\frac{p_{\theta}^{\text{RL}}(\hat{y}|\mathbf{x})}{p^{\text{PT}}(\hat{y}|\mathbf{x})} \right) \right] \end{aligned}$$

~ KL Divergence:

Give a Penalty when the model moves too far from original one

Too Tricky & Complex!



• DPO

- *Direct Preference Optimization*

- Idea

- Start from RLHF's Reward Model (Left)
- Can Calculate the Optimized Model

$$p^*(y|x) = \frac{p_{\text{ref}}(y|x) \exp\left(\frac{r(x,y)}{\beta}\right)}{Z(x)}$$

- Re-define the Reward Model(RM)

$$r(x, y) = \beta \log \frac{p^*(y|x)}{p_{\text{ref}}(y|x)} + \beta \log Z(x)$$

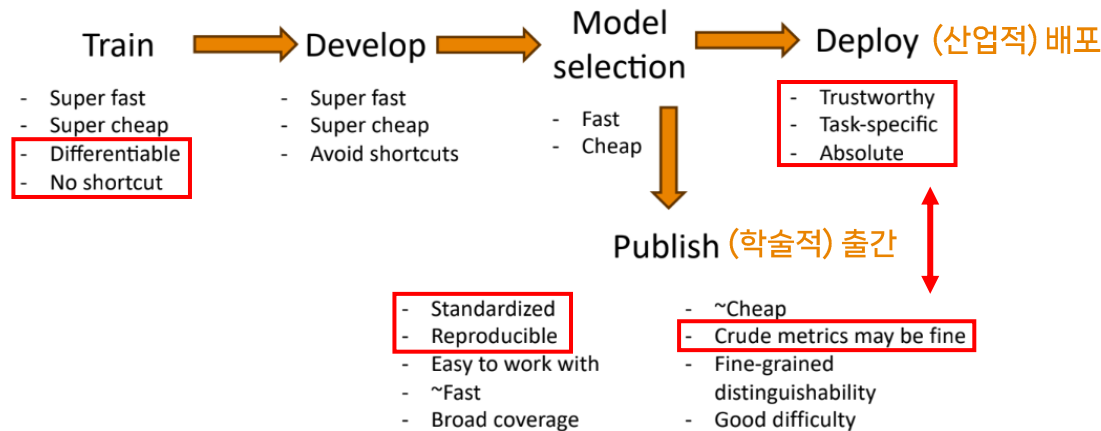
- Put the RM to Bradley-Terry Model(Loss)

$$-E_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{p_{\theta}(y_w|x)}{p_{\text{ref}}(y_w|x)} - \beta \log \frac{p_{\theta}(y_l|x)}{p_{\text{ref}}(y_l|x)} \right) \right]$$

⇒ No More Reinforcement Learning Problem

Lecture 11: Benchmarking & Evaluation

• Reasons for Measuring Performance



• Evaluation for Deploy

- Trustworthy
- Absolute

• Evaluation for Publish

- Reproducible
 - Cheap & Crude
 - *E.g., MMLU*
- ◀ Can be Imperfect, Show the Direction of Progress

• Two Types of Evaluations

- Close-Ended
 - *E.g., Text Classification*
- Open-Ended
 - *E.g., Text Generation*

Example

Text: Read the book, forget the movie!

Label: Negative

There is an Answer

Context (human-written): In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.

GPT-2: The scientist named the population, after their distinctive horn, Ovid's Unicorn. These four-horned, silver-white unicorns were previously unknown to science.

Now, after almost two centuries, the mystery of what sparked this odd phenomenon is finally solved.

Dr. Jorge Pérez, an evolutionary biologist from the University of La Paz, and several companions, were exploring the Andes Mountains when they found a small valley, with no other animals or humans. Pérez noticed that the valley had what appeared to be a natural fountain, surrounded by two peaks of rock and silver snow.

There is No 'One' Answer

Lecture 11

- Close-ended Problem

- Characteristics

- Limited # of Answers
 - One or Few Correct Answers
 - Enable Automatic Evaluation

- Task-Benchmark

- Sentiment Analysis: SST/IMDB/Yelp
 - Entailment: SNLI/CB/RTE
 - Part-of-Speech: PTB
 - Name Entity Recognition: CoNLL
 - Coreference Resolution: WSC
 - Question Answering: Squad 2

- Multi-task Benchmark

- SuperGLUE

- Challenges

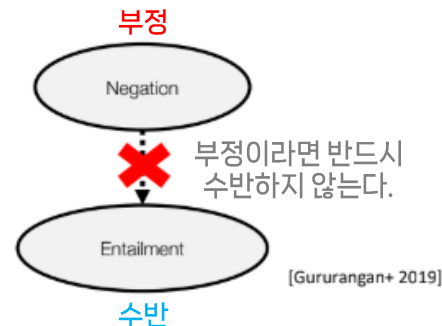
- Proper Metrics
 - Aggregate Metrics
 - Gathering Labels
 - Spurious Correlation

Premise: 전제

The economy could be still better.

Hypothesis: 가설

The economy has **never** been better



Model is Focusing on the Shortcut words (like Never, Not).

Don't do Reasoning Properly, Just Make its Answer by relying on the Shortcut words!

Lecture 11

- Open-ended Problem

- Characteristics

- Existence of Many Possible Correct Answers
 - Better & Worse Answer

- Tasks-Benchmark

- Summarization: CNN-DM/Gigaword
 - Translation: WMT
 - Instruction-following: Chatbot Arena

- Automatic Evaluations

- Content Overlap Metrics

- Compare Lexical Similarity between Generated and Gold-standard one
 - Fast & Efficient
 - E.g., BLEU(Precision), ROGUE(Recall)

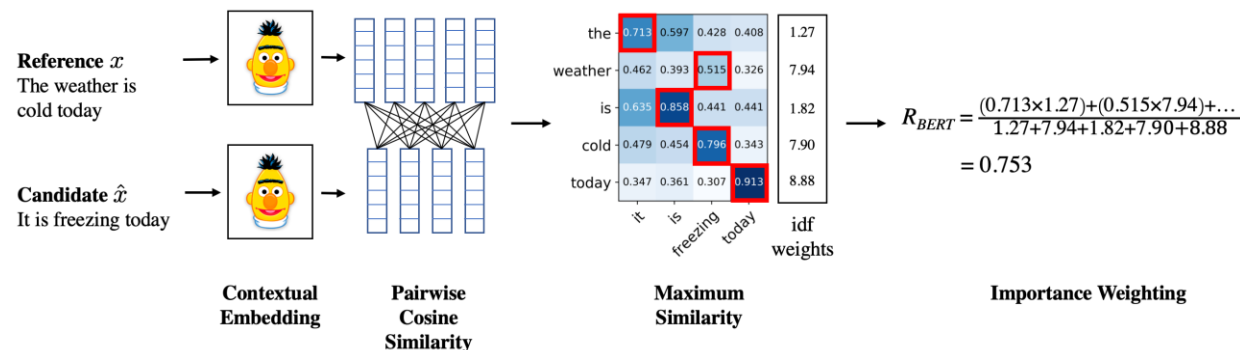
- Model-based Metrics

- Word Distance Functions

- BERTSCORE
 - BLEURT(BLEU+BERT)

- Reference-free Evaluations

- AlpacaEval
 - MT-Bench



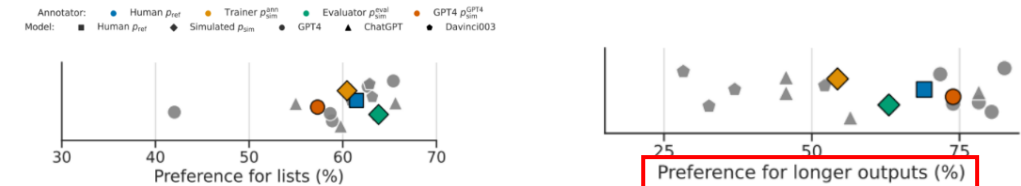
Lecture 11

• Human Evaluations

- Characteristics
 - Most Important Form of Eval
 - Gold Standard in Developing Metrics
- Problems
 - Can't Compare Scores
 - Slow & Expensive
 - Disagreement
 - Not Reproducible
 - Biases & Shortcuts
 - ...And Lots of Challenges
- Reference-free Eval
 - Chatbot Arena
 - LLM Evaluator

• LLM Evaluator

- Pros & Cons
 - ⇒ Cheaper & Faster
 - ⇒ Low Variance, High Bias
 - ⇒ Spurious Correlation Problems



• AlpacaEval

	AlpacaEval			Length-controlled AlpacaEval		
	concise	standard	verbose	concise	standard	verbose
gpt4_1106_preview	22.9	50.0	64.3	41.9	50.0	51.6
Mixtral-8x7B-Instruct-v0.1	13.7	18.3	24.6	23.0	23.7	23.2
gpt4_0613	9.4	15.8	23.2	21.6	30.2	33.8
claude-2.1	9.2	15.7	24.4	18.2	25.3	30.3
gpt-3.5-turbo-1106	7.4	9.2	12.8	15.8	19.3	22.0
alpaca-7b	2.0	2.6	2.9	4.5	5.9	6.8

Lecture 11

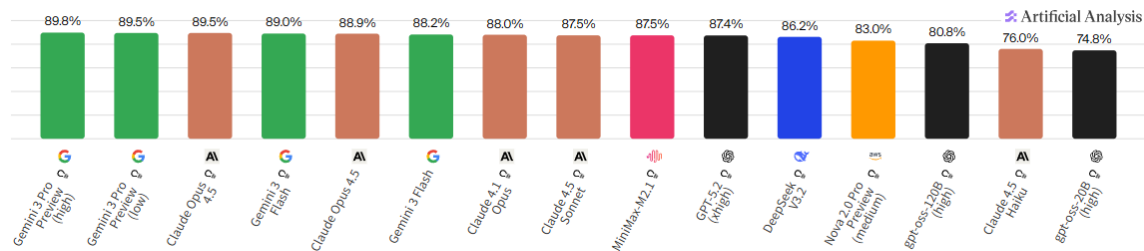
- Evaluation of LLM

- Currently?

- Perplexity
 - Everything
 - HELM
 - Huggingface Leaderboard
 - Arena-like

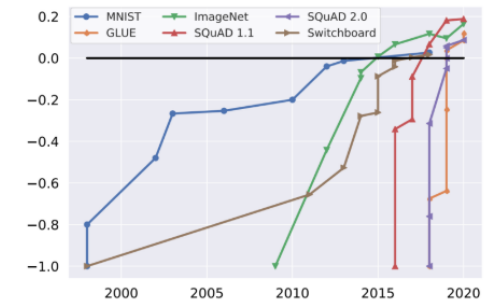
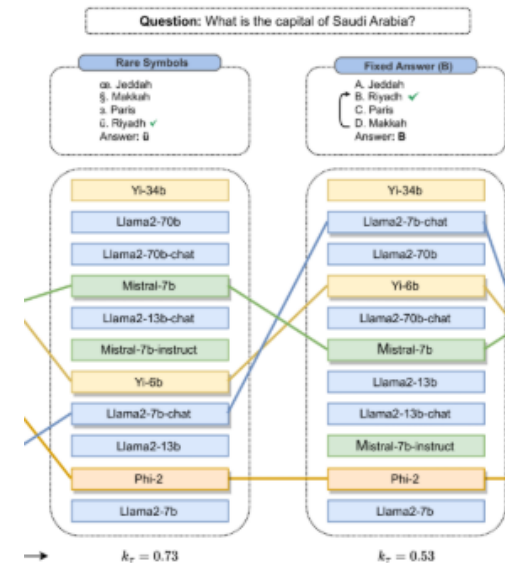
- E.g., MMLU/MMLU-Pro

- *Massive Multitask Language Understanding*
 - 57 diverse knowledge intensive tasks



- Issue & Challenge

- Consistency Issues
 - Contamination
 - Overfitting
 - Monoculture

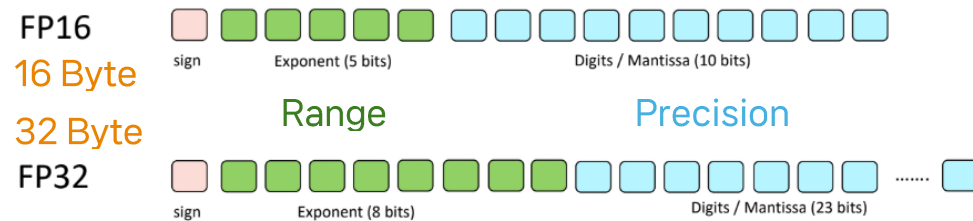


Reach "human-level" performance too quickly

Lecture 12: Efficient Training

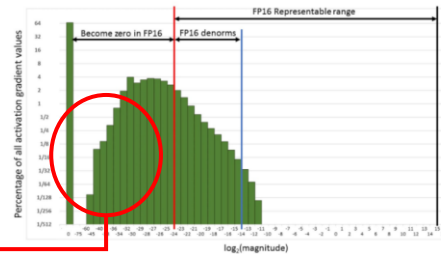
• Mixed Precision Training

- Floating Points



• FP32 → FP16

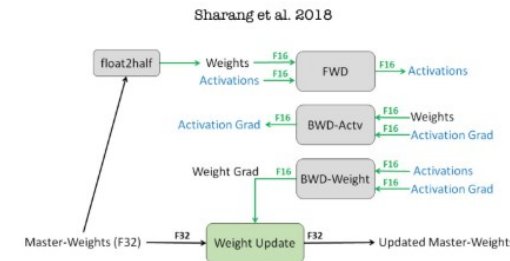
- Pros
 - Low Memory Usage
 - Solution of OOM
- Cons
 - Cause Underflow
 - Imprecise Update



Half of Gradients go to Zero
→ Underflow!

• Mixed Precision Training

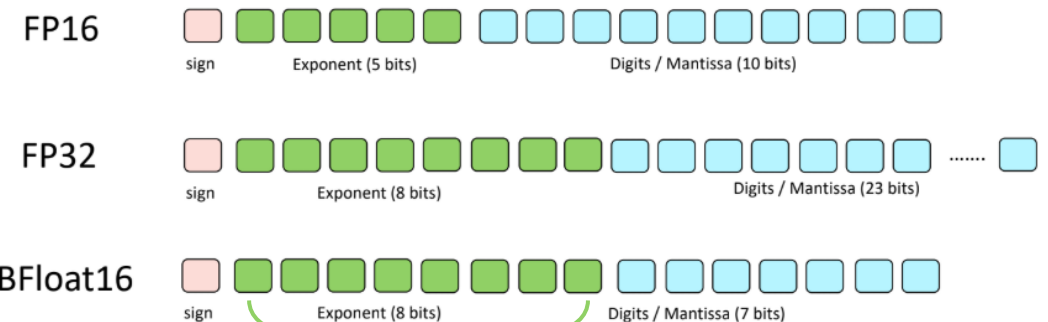
- Idea
- Gradient Scaling



Recipe for Mixed-Precision Training

1. Maintain a copy of model parameters in FP32 (Master weights)
2. Run forward pass in FP16
3. Scale loss by a large value (to artificially increase gradient)
4. Compute gradient in FP16
5. Copy gradient into FP32 and divide by scale factor
6. Update master weights in FP32 [fixes weight update issue!]
7. Copy into FP16 version

• BFloat16



Attention Is All You Need (2017)

- Paper Review (30m)
 - Intro & Background
 - Model Architecture
 - Self-Attention
 - Training
 - Results & Conclusion
- Implementation(10m)
 - Annotated Transformer
- Train & Test(10m)
 - Train Process
 - Test Result
- Intro & Background
- Model Architecture •
 - Encoder & Decoder
 - Attention
 - Position Wise FFN
 - Embeddings & SoftMax
 - Positional Encoding
- Self-Attention •
- Training
- Results & Conclusion

Paper Review: Intro & Background

• Introduction

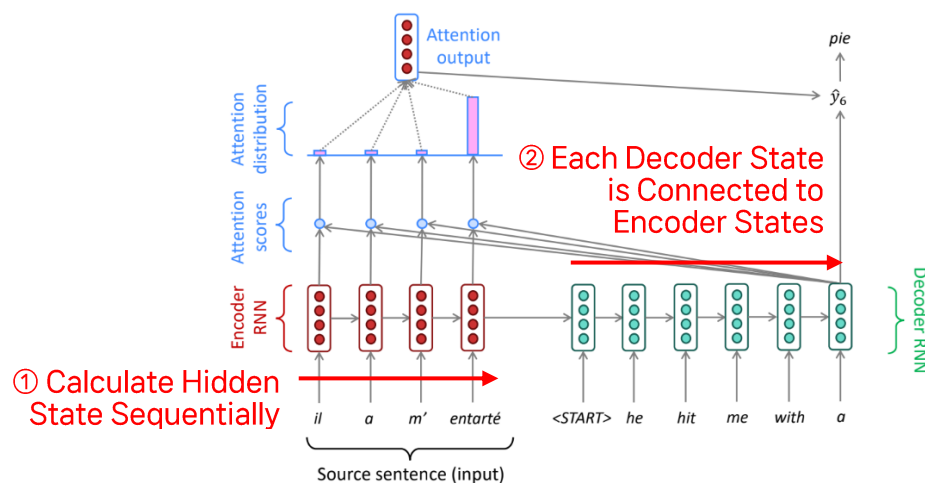
• Limitations of RNN Architecture

- Hidden States Need to Calculate Sequentially

→ Precludes Parallelization

- Decoder-Encoder Attention

→ Dependency on Time Steps

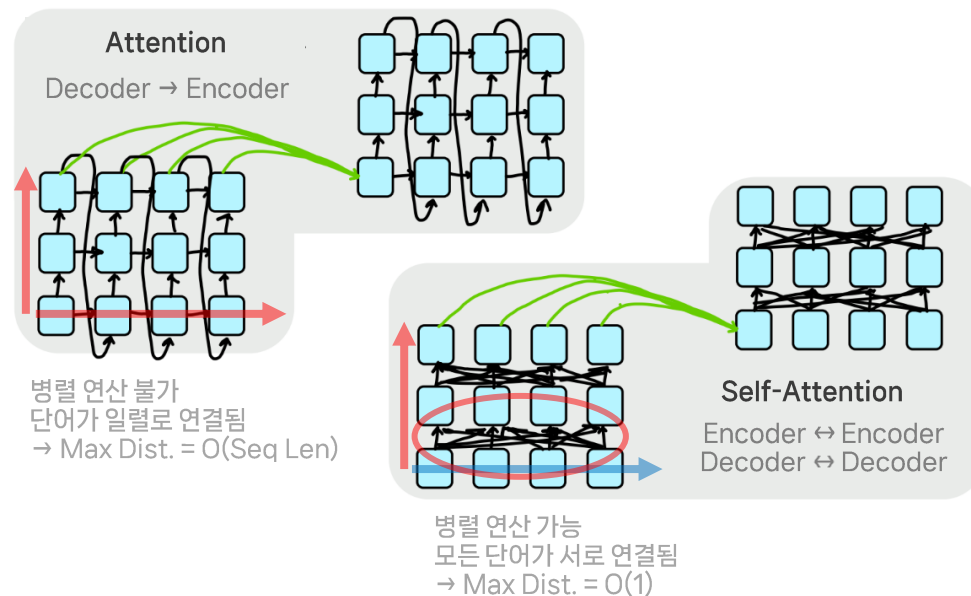


• Background

• Compute States in Parallel

- Limitations of CNN based Architecture

• Self-Attention Mechanism



Paper Review: Model Architecture

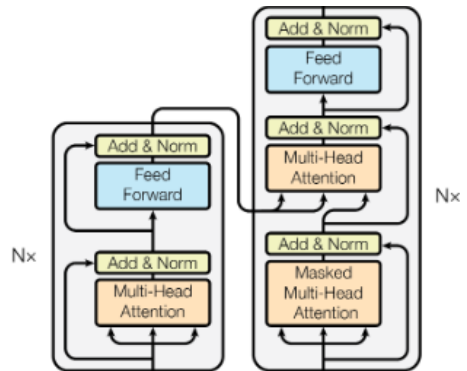
- Encoder & Decoder

- Common Structure

- Num of Identical Layers (N=6)
 - Multi-Head Attention
 - Attention Layer → FFN Process
 - Residual Connection + Layer Norm

- Difference in Decoder

- Num of Attention Layers (1 → 2)
 - Masking Multi Head Attention



- Attention

Normalized Attention Score(A)
Energy or Attention Score(E)

- $Output = \text{softmax}(QK^T)V$

- Scaled Dot Product Attention

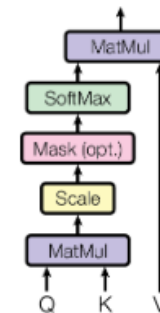
$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

- Multi-Head Attention

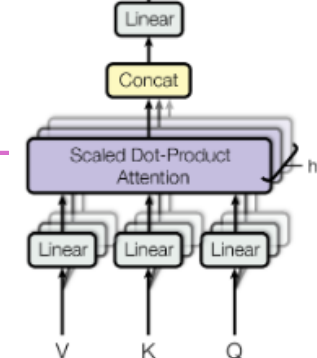
$$MultiHead(Q, K, V) = \text{Concat}(head_1, \dots, head_h)W^O$$

- Applications

Scaled Dot-Product Attention



Multi-Head Attention



Paper Review: Model Architecture

- Position Wise Feed Forward Network

- Two Linear Functions
- ReLU Activation Function
- $FFN(x) = \max(0, xW_1 + b_1)W_2 + b_2$

- Embeddings & SoftMax

- Embedding Size = d_{model} (512)
- Weight Sharing

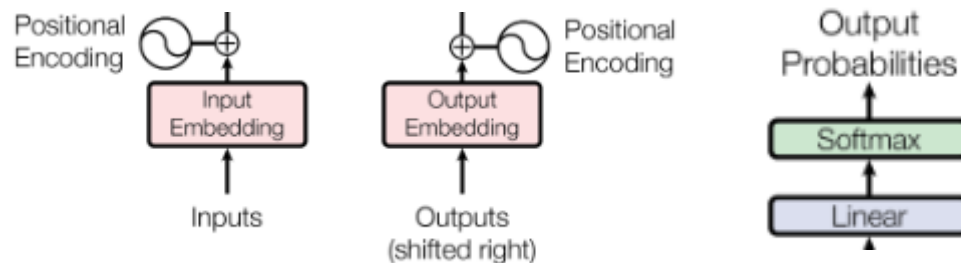
- Positional Encodings

- Use Sinusoidal Function

- pos = Position in (Input) Sequence
- i = Index of Dimension

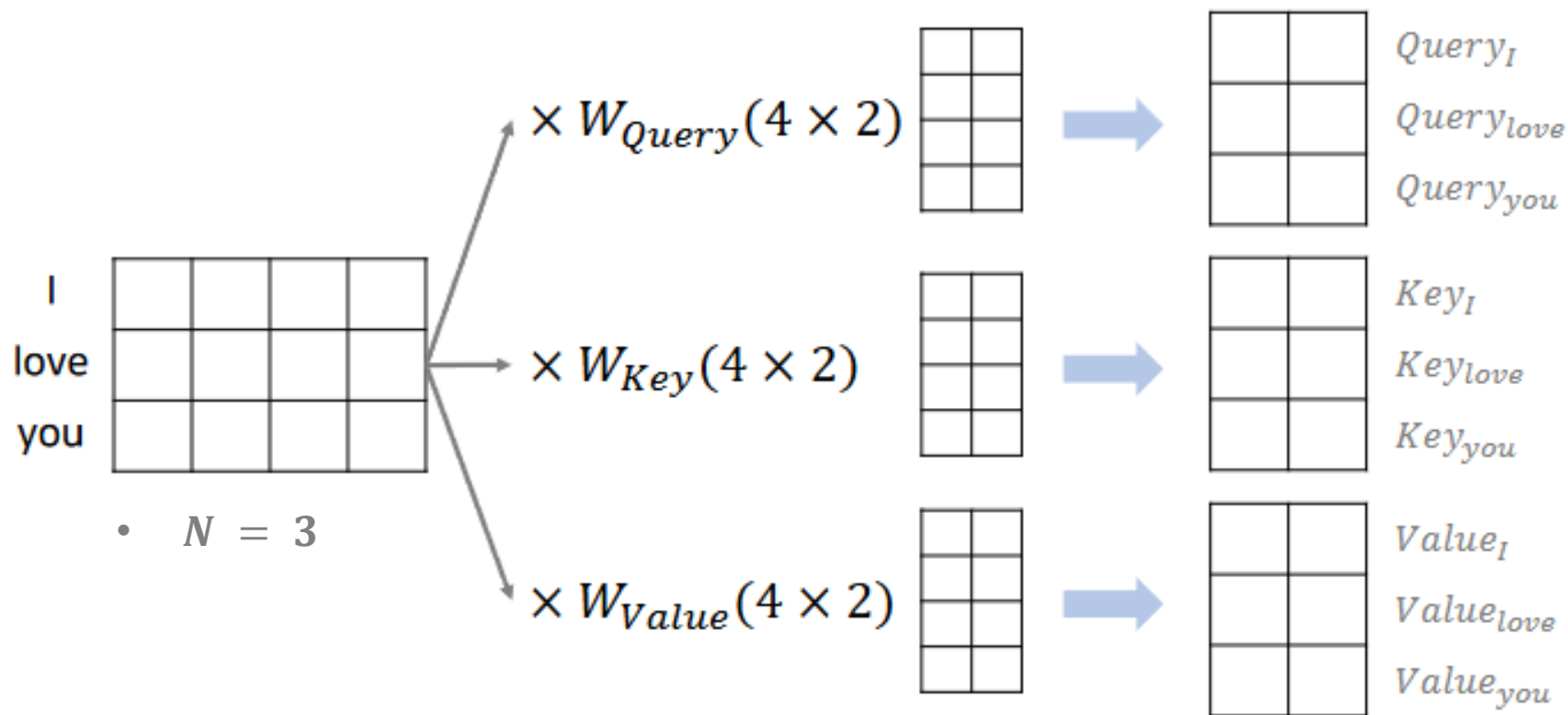
$$PE_{(pos, 2i)} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

$$PE_{(pos, 2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$



Paper Review: Model Architecture

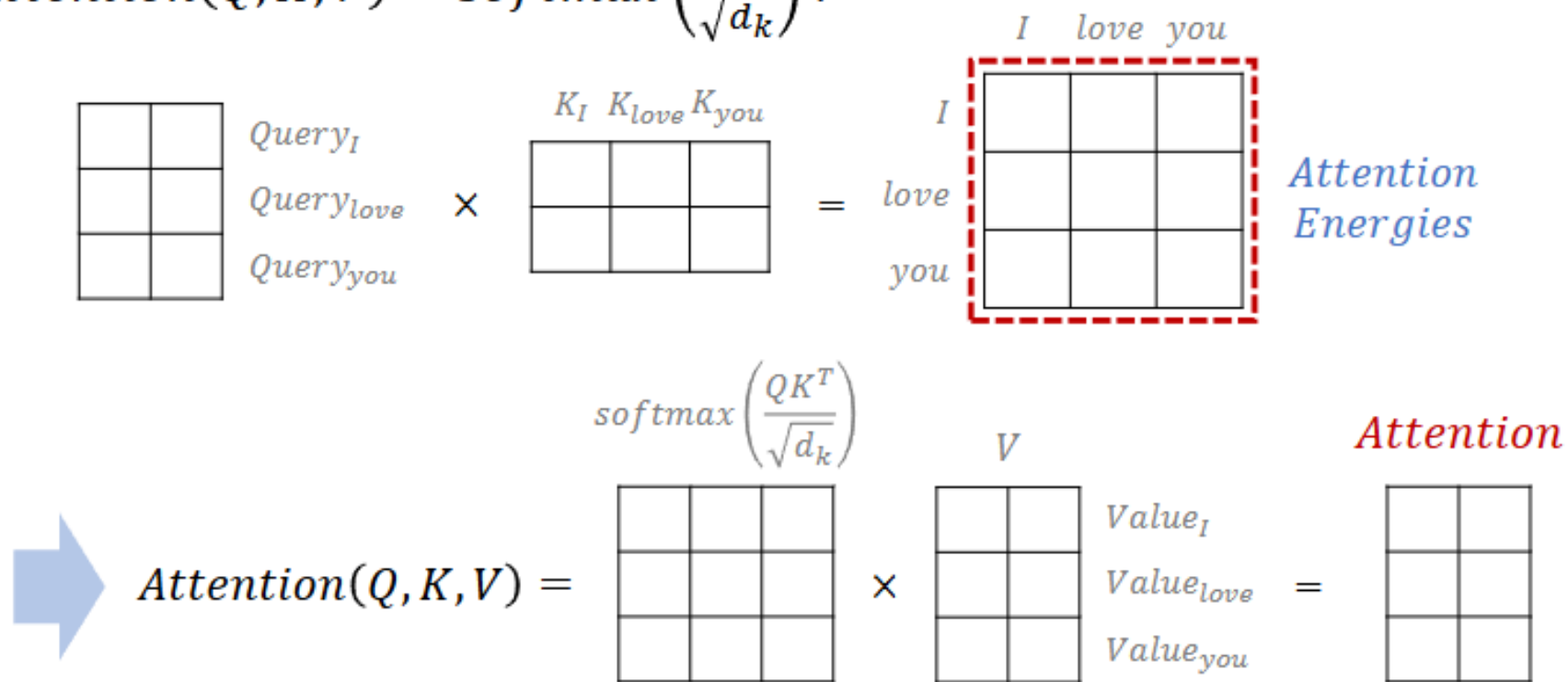
- **Overall Process** (Original Slides: [Dongbin Na](#))



Paper Review: Model Architecture

- **Overall Process** (Original Slides: [Dongbin Na](#))

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$



Paper Review: Self Attention

- Self-Attention
- Benefits of Self Attention
 - Complexity per Layer ↓
 - Parallelized Operations ↑
 - Path Length Learnability ↑

Layer Type	Complexity per Layer	Sequential Operations	Maximum Path Length
Self-Attention	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$
Convolutional	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k(n))$
Self-Attention (restricted)	$O(r \cdot n \cdot d)$	$O(1)$	$O(n/r)$

n = (Input) Sequence Length

d = Model Dimension

k = kernel Size

Paper Review: Training

- Training Data

- WMT 2014
 - English-German(4.5M)
 - English-French(36M)

- Hardware

- 8 NVIDIA P100 GPUs
- Schedule
 - Base: 100,000 steps (12H)
 - Big: 300,000 steps (3.5D)

- Optimizer

- Adam
- Flexible LR

$$lr_{rate} = d_{model}^{-0.5} \cdot \min(step_num^{-0.5}, step_num \cdot warmup_steps^{-1.5})$$

- Regularization

- Residual Dropout
- Label Smoothing
 - Perplexity ↓
 - BLEU Score ↑

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	41.29	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	$3.3 \cdot 10^{18}$	
Transformer (big)	28.4	41.8	$2.3 \cdot 10^{19}$	

Paper Review: Results & Conclusion

• Results

- Machine Translation
- Model Variations
- English Constituency Parsing

• Conclusion

- Attention Only Architecture
No CNN or RNN
- Fast & SoTA Performance
- Potential for Scaling

	N	d_{model}	d_{ff}	h	d_k	d_v	P_{drop}	ϵ_{ls}	train steps	PPL (dev)	BLEU (dev)	params $\times 10^6$	
base	6	512	2048	8	64	64	0.1	0.1	100K	4.92	25.8	65	
(A)					1	512	512				5.29	24.9	
					4	128	128				5.00	25.5	
					16	32	32				4.91	25.8	
					32	16	16				5.01	25.4	
(B)					16					5.16	25.1	58	
					32					5.01	25.4	60	
(C)	2									6.11	23.7	36	
	4									5.19	25.3	50	
	8									4.88	25.5	80	
	256				32	32				5.75	24.5	28	
	1024				128	128				4.66	26.0	168	
			1024							5.12	25.4	53	
			4096							4.75	26.2	90	
(D)							0.0			5.77	24.6		
							0.2			4.95	25.5		
								0.0		4.67	25.3		
								0.2		5.47	25.7		
(E)	positional embedding instead of sinusoids									4.92	25.7		
big	6	1024	4096	16				0.3	300K	4.33	26.4	213	